

SADMAN SAKIB

Lecturer

Department of Electrical and Electronic Engineering
Metropolitan University

Email: ssakib12325@gmail.com ; sadmansakib92@iut-dhaka.edu

Website: ssakib92.github.io

LinkedIn: [Sadman Sakib](#) ; Phone: +8801868889767

Education

Bachelor of Science (B.Sc.) in Electrical & Electronic Engineering

Islamic University of Technology (IUT)

Jan 2018 – April 2022

CGPA: 3.71 / 4.00 (**Last 4 Semesters: 3.87**)

Research Interests

- Neuromorphic Computing and Brain-Inspired Systems
- Spintronic and Emerging Memory Devices (MTJ, 2D Materials)
- Software-Hardware Co-Design and Inverse Design
- Applied Machine Learning for Physical Systems

Professional Experience

Lecturer, Department of Electrical and Electronic Engineering

Metropolitan University, Sylhet

August 2023 – Present (Full-time)

Courses Instructed:

- CSE 123: Basic Electrical Engineering
- CSE 211: Digital Logic Design

Technology Program Management Intern

Banglalink Digital Communication Limited

May 2022 – August 2022 (Full-time)

- Worked with the program management team on various projects, assisting in day-to-day operations.

Research and Publications

Undergrad Thesis:

Title: Hybrid Algorithm-Based Approach for Optimized PID Controller for Stability Analysis of DC-DC Interleaved-Buck Converter

Objective: The research focuses on analyzing the stability analysis of the Interleaved Buck converter leveraging Nature-Inspired Optimization Algorithms to design an optimal PID controller. The thesis aims to optimize performance parameters, minimize ripple currents, achieve faster transient responses, and assess results using various fitness functions, demonstrating the effectiveness of Nature-Inspired Algorithms in establishing robust and efficient feedback control for power converters.

Area: Power Electronics, Converters, Optimization, Control

Supervisor: Mr. Mirza Muntasir Nishat

Publications:

1. Attention-Based Models for Multivariate Time Series Forecasting: Multi-step Solar Irradiation Prediction

Published in Heliyon, Volume 10, Issue 6, 2024 [[link](#)]

Authors: Sadman Sakib, Mahin K. Mahadi, Samiur R. Abir, *et al.*

2. Applying Particle Swarm Optimization for Investigating the Stability of DC-DC Buck-Boost Converter: A Software Based Approach

Published in International Journal of Modelling and Simulation, 2024 [\[link\]](#)

Authors: Mirza M. Nishat, Sadman Sakib, Kazi Asif Rahman, *et al.*

3. Surface plasmon resonance–based biosensor for accurate and rapid detection of Mycobacterium tuberculosis

Published in Microchimica Acta (Springer), 2025 [\[link\]](#)

Authors: Ahmed Aff Rafsan, Simanta Das, Kowshik Kumar Roy, Sadman Sakib, *et al.*

Ongoing Project

- **MTJ-Based True Random Number Generator (TRNG) Design**

Stochastic MTJ modeling and optimization using evolutionary algorithm techniques for True Random Number Generator (TRNG) application.

- **DFT Study of 2D Material**

Exploring the strain-dependent properties of monolayer 2D material using first-principles simulations using *Quantum ESPRESSO* to evaluate its potential for scalable device applications.

Supervisor: Dr. Md. Mohsinur Rahman Adnan, Assistant Professor, Bangladesh University of Engineering and Technology (BUET)

Affiliations and Volunteering Experience

IUT Mars Rover – Team Avijatrik

Team Lead, Mechanical Design

Designed and led the development of a Mars rover prototype for international competitions; responsible for overall mechanical system design and integration.

Achievement: European Rover Challenge 2019 (Poland) — *15th Place overall, 2nd in Asia.*

IEEE Robotics & Automation Society, IUT Student Branch

Vice Chair

Organized technical workshops and webinars on robotics, MATLAB, and PCB design; coordinated activities with multiple university clubs.

IUT CAD Society

Chief of Management

Planned and managed workshops and training sessions on AutoCAD and Fusion360.

Skills

Programming: Python, MATLAB, C, C++

ML & Scientific Computing: PyTorch, NumPy, Pandas, Matplotlib

Simulation & Design Tools: MATLAB Simulink, Quantum ESPRESSO, PSpice, Proteus

Research Skills: Scientific Writing, Statistical Analysis, Data Visualization

Test Scores

IELTS Academic Score: Overall 7.5/9.0

(Listening: 8.5/9.0, Reading: 7.5/9.0, Writing: 6.5/9.0, Speaking: 6.5/9.0) – CEFR Level: C1

References

Mr. Mirza Muntasir Nishat

Assistant Professor

Islamic University of Technology

Tel: +8801676151945

Email: eee.mirzamuntasir@iut-dhaka.edu

Mr. Nawshad Ahmed Chowdhury

Assistant Professor

Head, Dept. of EEE, Metropolitan University

Tel: +8801670260481

Email: nawshad@metrouni.edu.bd